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Kawamata et al.

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(54) **ULTRASOUND PROBE AND ULTRASOUND DIAGNOSTIC APPARATUS**

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(30) **Foreign Application Priority Data**

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Oct. 6, 2023 (JP) 2023-174159

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G01S 15/89 (2006.01)

(52) **U.S. Cl.**
CPC ***A61B 8/4488*** (2013.01); ***G01S 15/8925*** (2013.01)

(58) **Field of Classification Search**
CPC A61B 8/4488; A61B 8/4494; A61B 8/54; G01S 15/8925; G01S 15/8915; G01S 15/8927
See application file for complete search history.

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(57) **ABSTRACT**

The number of signal lines connecting an ultrasound probe and an apparatus main body of an ultrasound diagnostic apparatus can be reduced while suppressing reduction of the maximum opening width of the ultrasound probe. A transducer array is conceptually divided into a plurality of center-side main transducers located on a center side of an arrangement, a plurality of first end-side main transducers located on a first end side of the arrangement with respect to the plurality of center-side main transducers, and a plurality of second end-side main transducers located on a second end side with respect to the plurality of center-side main transducers. A connection circuit is configured to include a first circuit connecting a transmission and reception circuit and the center-side main transducer, and a second circuit provided with a switch as a main transducer switching switch for switching a connection destination of the transmission and reception circuit between the first end-side main transducer and the second end-side main transducer.

7 Claims, 9 Drawing Sheets

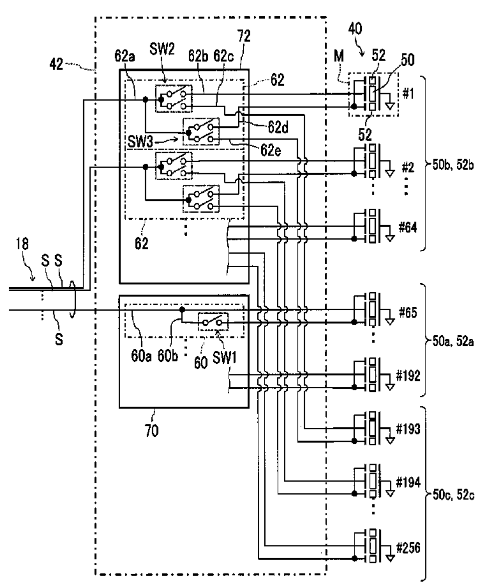


FIG. 1

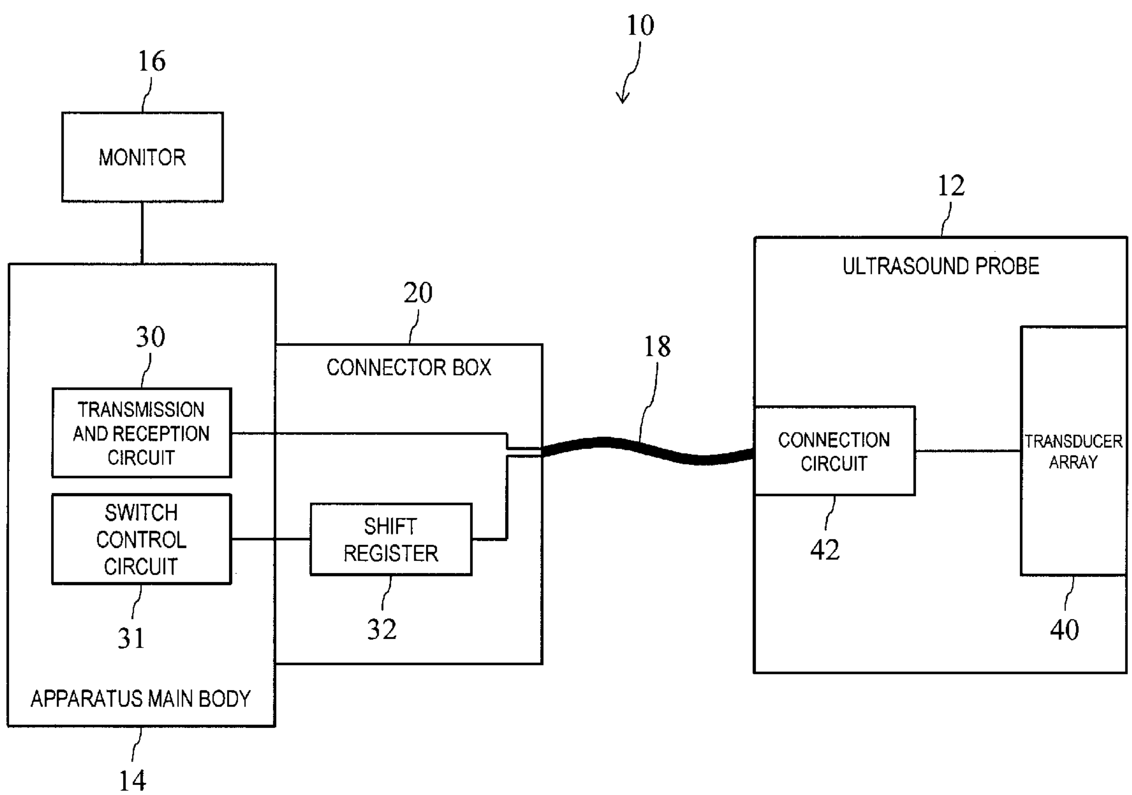

10
16
MONITOR
12
ULTRASOUND PROBE
20
CONNECTOR BOX
30
18
TRANSMISSION AND RECEPTION CIRCUIT
CONNECTION CIRCUIT
TRANSDUCER ARRAY
SWITCH CONTROL CIRCUIT
SHIFT REGISTER
42
31
32
40
APPARATUS MAIN BODY
14

FIG. 2

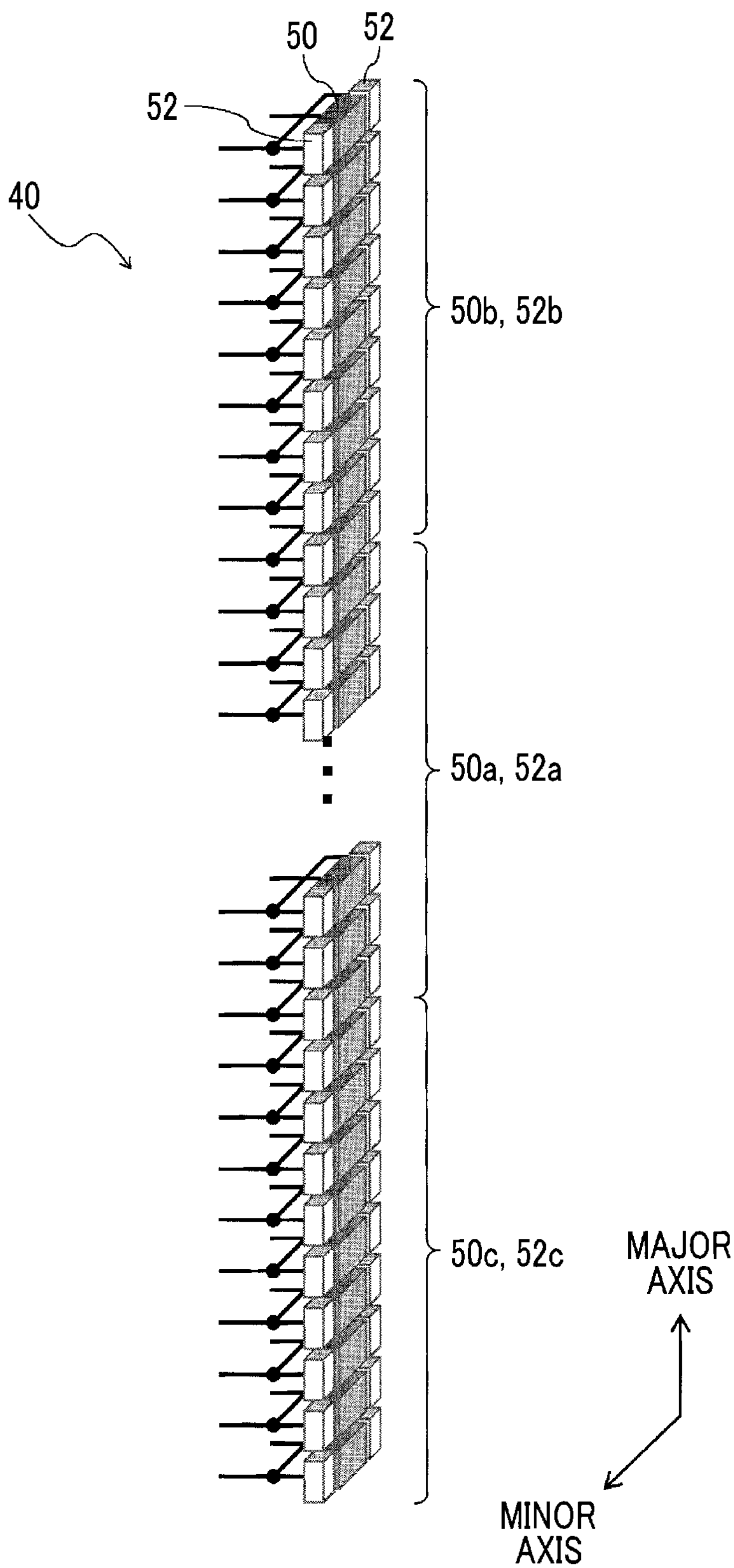
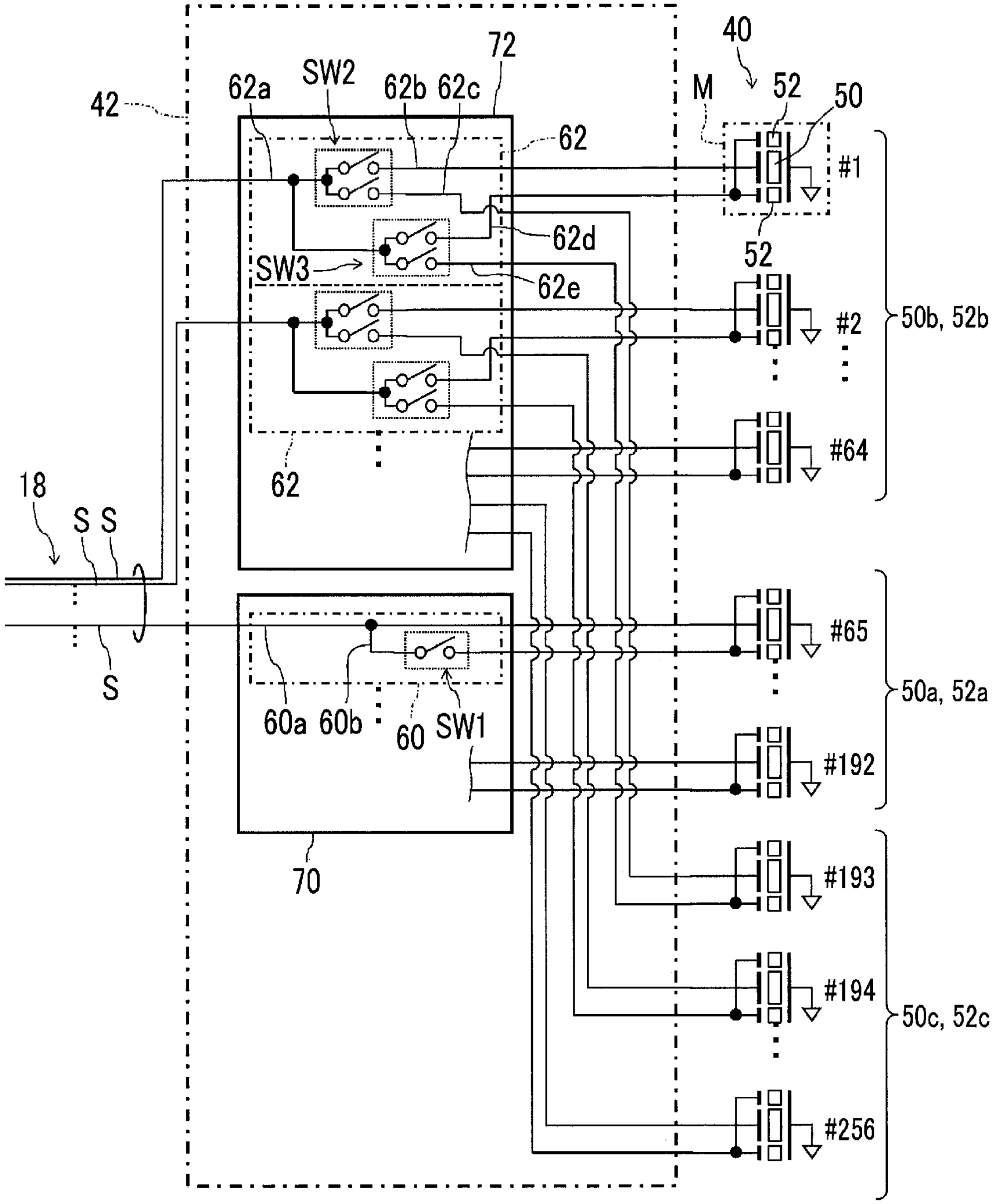


FIG. 3



42
72
40
M
52
50
62a
SW2
62b
62c
62
#1
62d
52
SW3
62e
#2
50b, 52b
#64
62
18
S
S
#65
S
60a
60b
60
SW1
50a, 52a
#192
70
#193
#194
50c, 52c
#256

FIG. 4

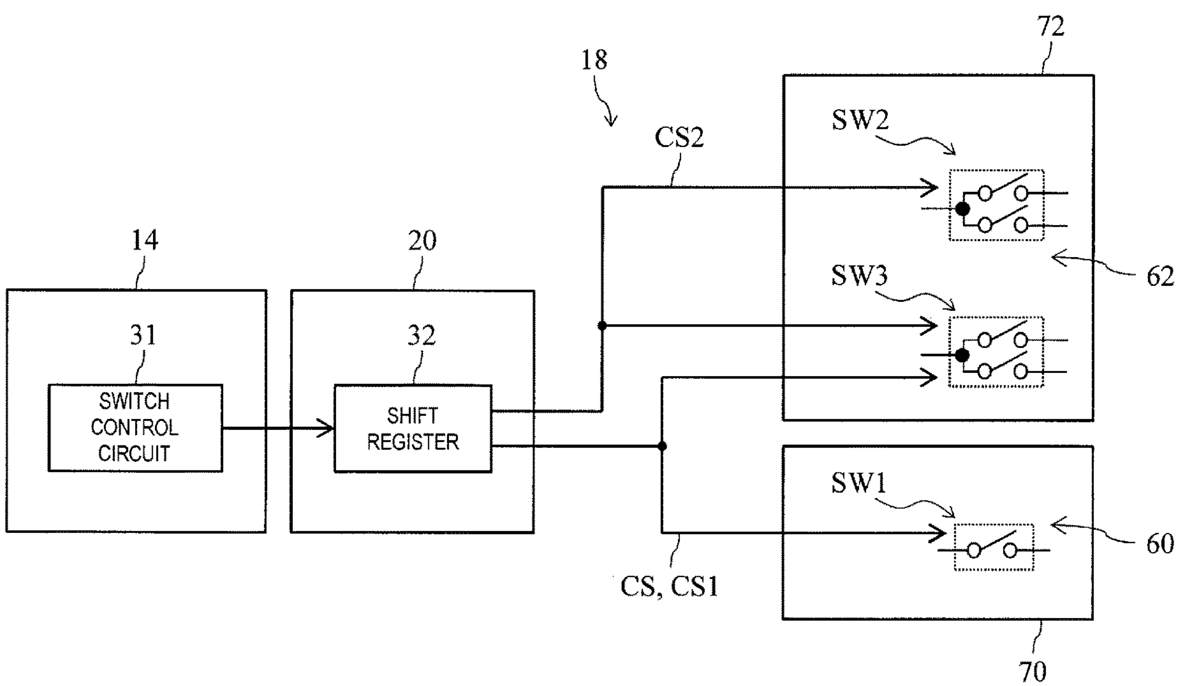

18
72
CS2
SW2
14
20
62
31
32
SW3
SWITCH
CONTROL
CIRCUIT
SHIFT
REGISTER
SW1
60
CS, CS1
70

FIG. 5

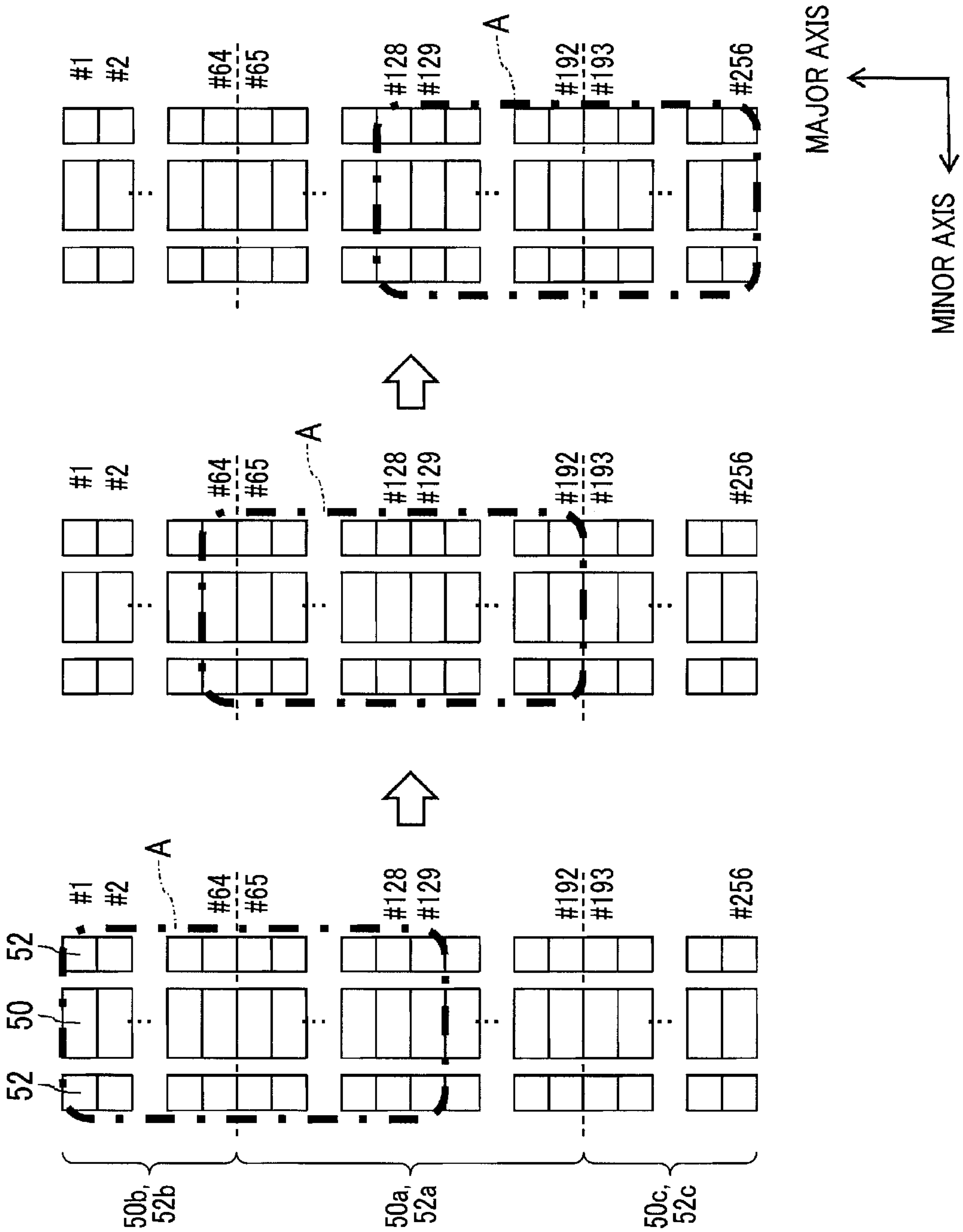

A
#1
#2
#64
#65
#128
#129
#192
#193
#256
52
50
52
50b, 52b
50a, 52a
50c, 52c
MAJOR AXIS
MINOR AXIS

FIG. 6

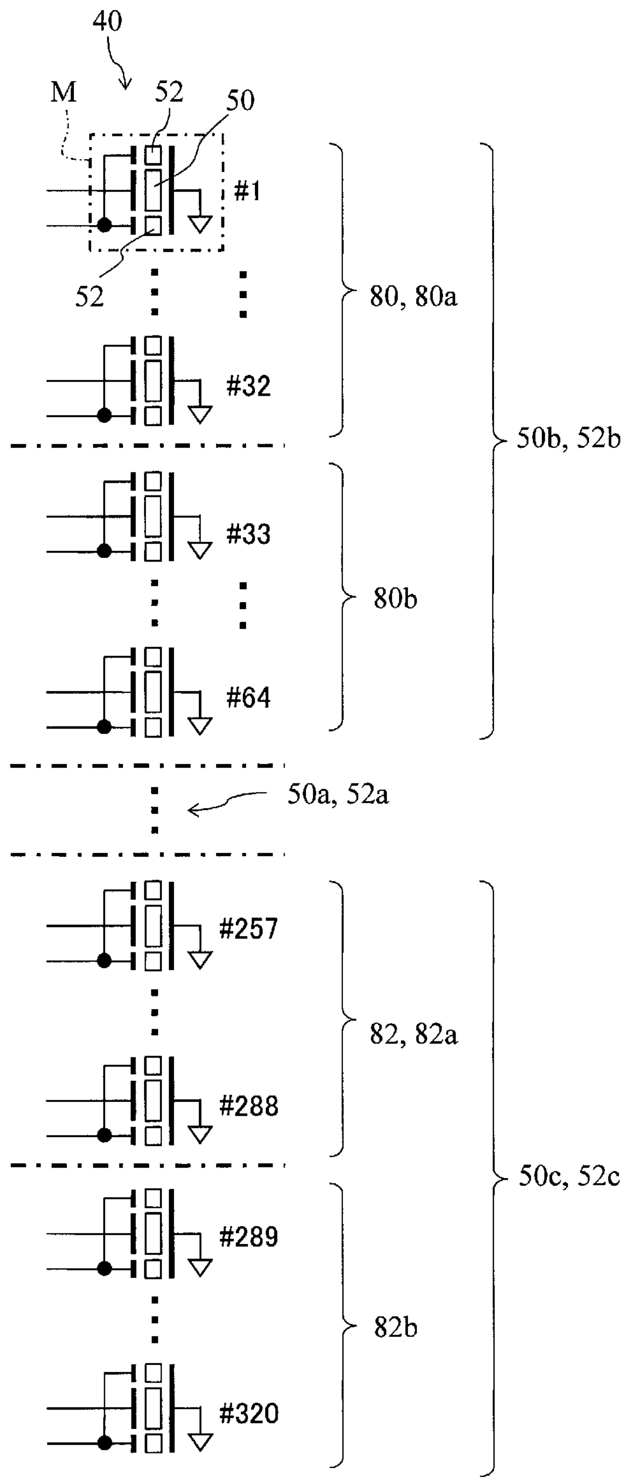
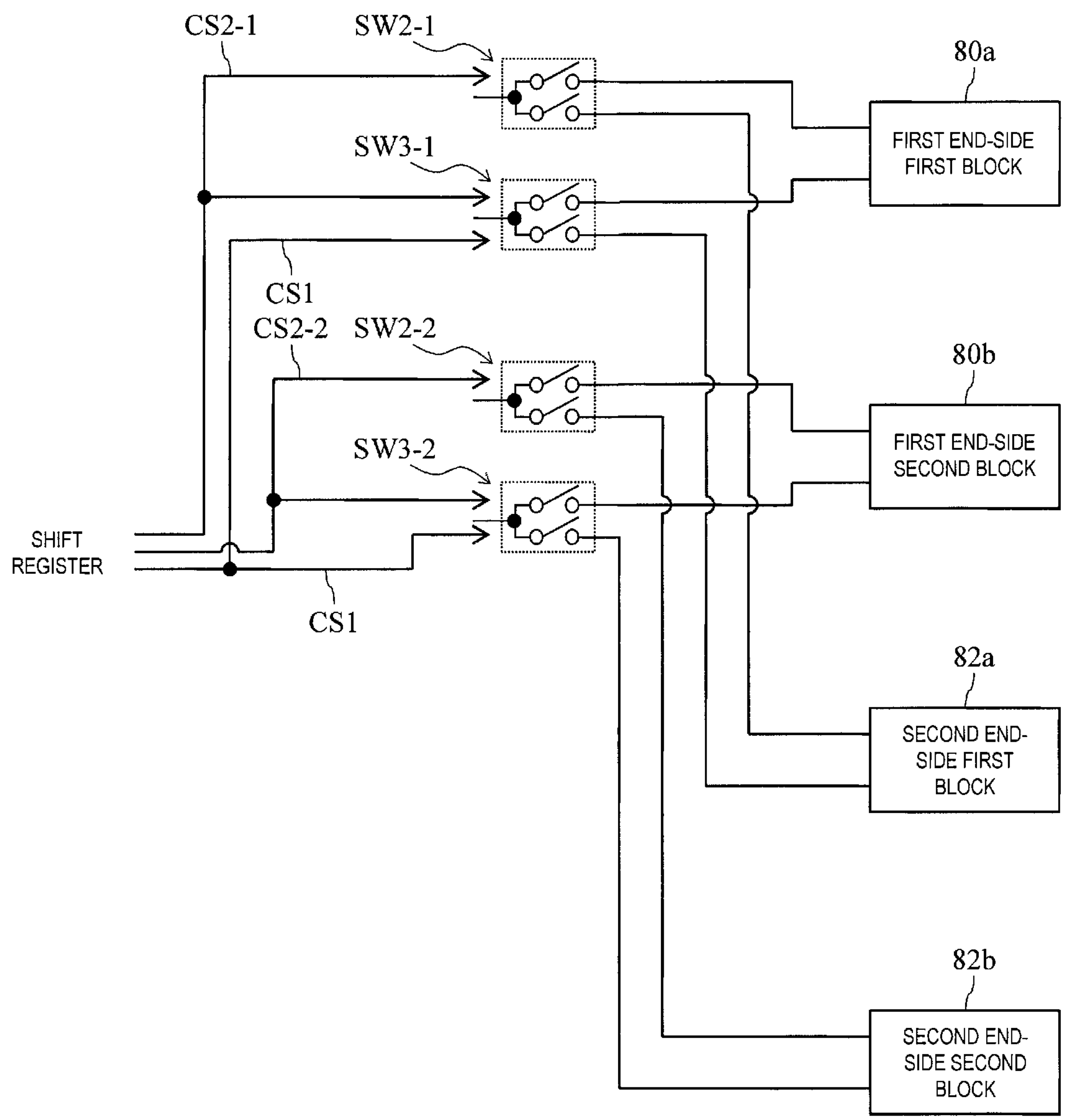

40
M
52
50
#1
52
80, 80a
#32
50b, 52b
#33
80b
#64
50a, 52a
#257
82, 82a
#288
50c, 52c
#289
82b
#320

FIG. 7



CS2-1
SW2-1
80a
FIRST END-SIDE FIRST BLOCK
SW3-1
CS1
CS2-2
SW2-2
80b
FIRST END-SIDE SECOND BLOCK
SW3-2
SHIFT REGISTER
CS1
82a
SECOND END-SIDE FIRST BLOCK
82b
SECOND END-SIDE SECOND BLOCK

FIG. 8

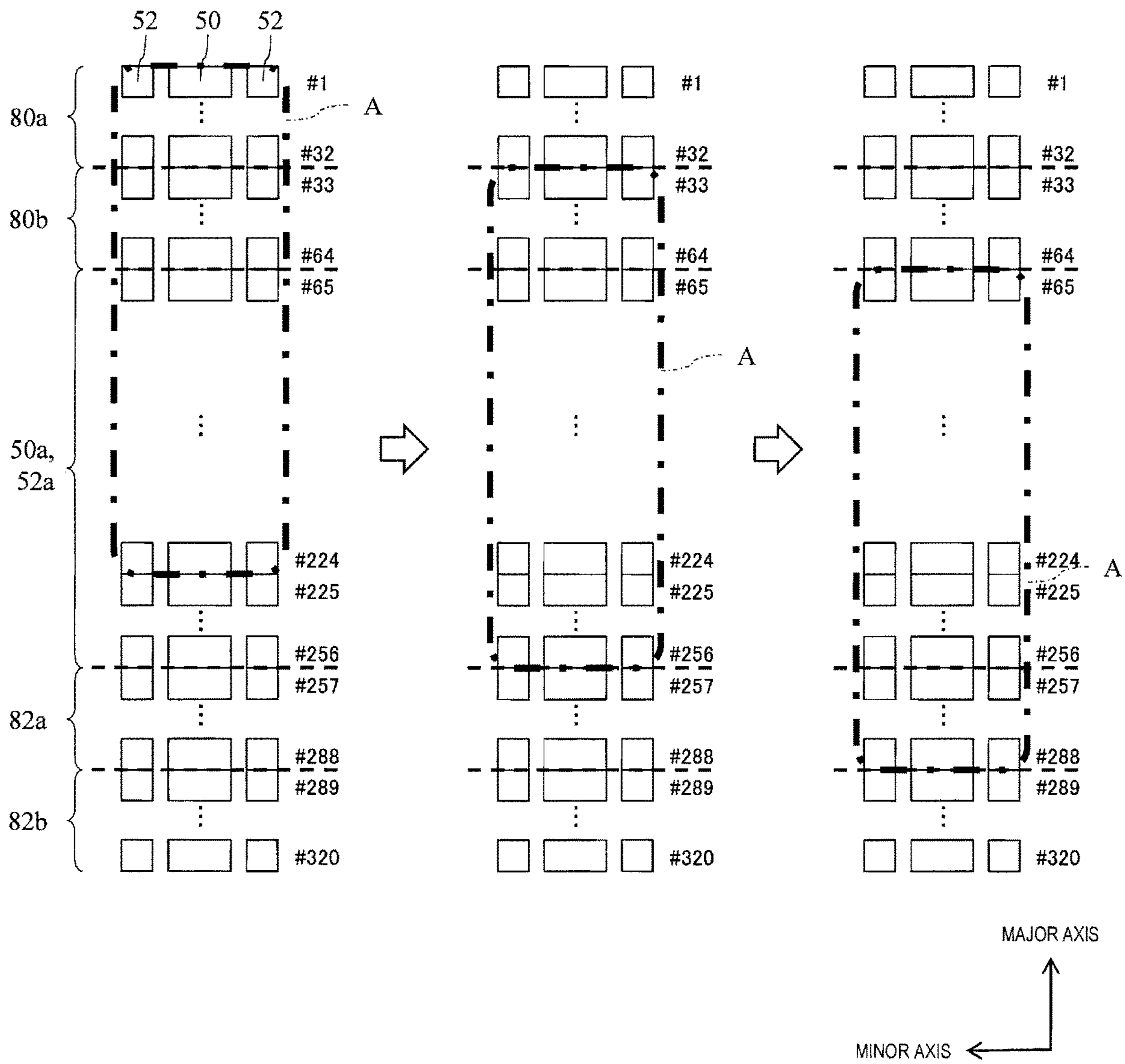
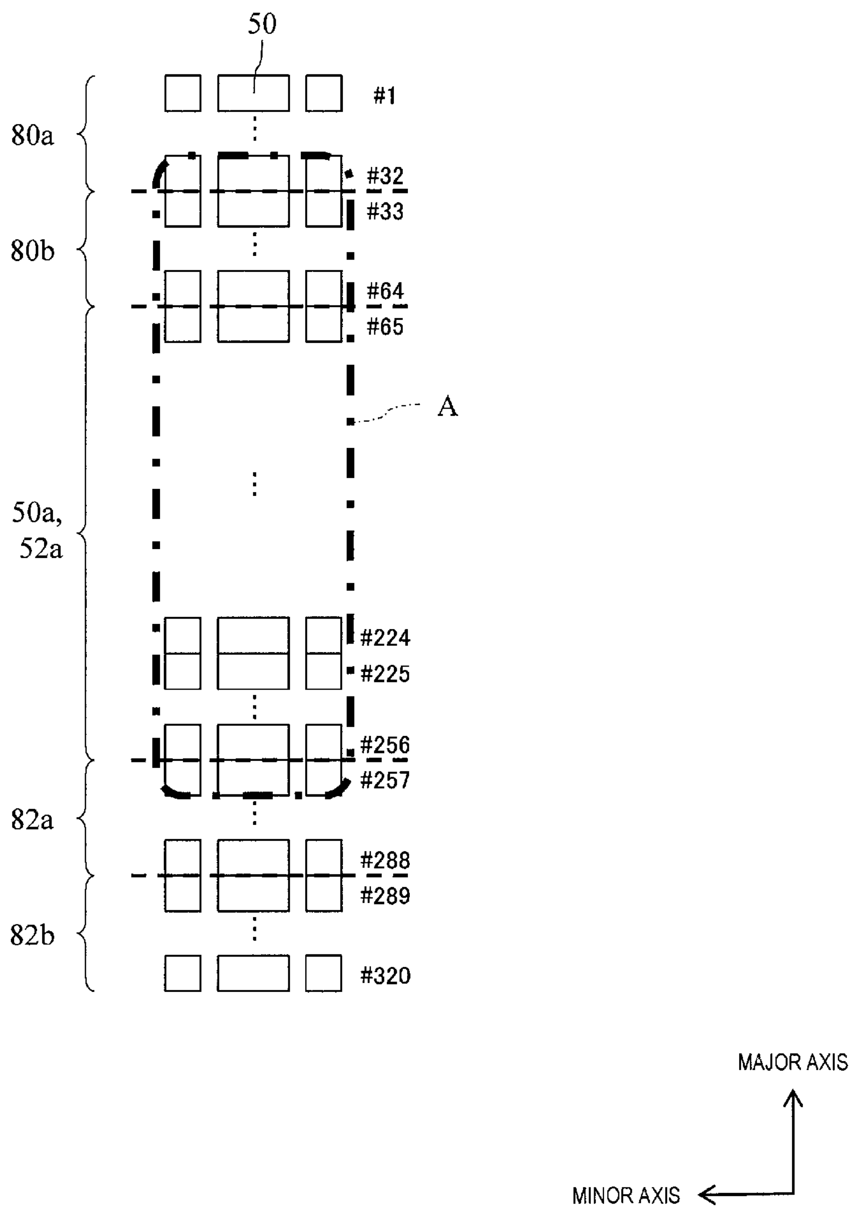

52
50
52
80a
80b
50a,
52a
82a
82b
A
#1
#32
#33
#64
#65
#224
#225
#256
#257
#288
#289
#320
MAJOR AXIS
MINOR AXIS

FIG. 9


50
#1
80a
#32
#33
80b
#64
#65
A
50a,
52a
#224
#225
#256
#257
82a
#288
#289
82b
#320
MAJOR AXIS
MINOR AXIS

ULTRASOUND PROBE AND ULTRASOUND DIAGNOSTIC APPARATUS

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application claims priority from Japanese Patent Applications Nos. 2022-180142 and 2023-174159 filed on Nov. 10, 2022 and Oct. 6, 2023, respectively, the content of each of which is hereby incorporated by reference into this application.

BACKGROUND OF THE INVENTION

1. Field of the Invention

The present specification discloses improvement of an ultrasound probe and improvement of an ultrasound diagnostic apparatus.

2. Description of the Related Art

In the related art, there has been known an ultrasound diagnostic apparatus that transmits and receives an ultrasonic wave to and from a subject, forms an ultrasound image based on a reception signal obtained by the transmission and reception of the ultrasonic wave, and displays the formed ultrasound image. In general, the ultrasound diagnostic apparatus is configured to include an ultrasound probe, an apparatus main body, and a monitor. The ultrasound probe transmits and receives an ultrasonic wave to and from a subject. The apparatus main body performs a process of transmitting a transmission signal for causing the ultrasound probe to transmit the ultrasonic wave, a process of forming an ultrasound image based on a reception signal from the ultrasound probe, or the like. The monitor displays the ultrasound image formed by the apparatus main body.

In recent years, a wireless probe has also been proposed, but there are still many cases where the apparatus main body and the ultrasound probe are connected via a cable. The ultrasound probe has a transducer array including a large number of transducers (for example, hundreds of transducers) for transmitting and receiving the ultrasonic wave. Therefore, in order for the apparatus main body to send the transmission signal to each transducer or for the apparatus main body to receive the reception signal from each transducer, the cable connecting the apparatus main body and the ultrasound probe has to have a large number of signal lines (generally, signal lines whose number corresponds to the number of the transducers).

Therefore, in the related art, a technique for reducing the number of the signal lines of the cable connecting the apparatus main body and the ultrasound probe has been proposed. For example, JP2012-152317A discloses an ultrasound probe including a plurality of switches that divide a transducer array of the ultrasound probe into an A group and a B group and switch a connection destination of a signal line from an apparatus main body to either a transducer in the A group or a transducer in the B group. According to this, one signal line from the apparatus main body can correspond to a plurality of transducers by the switching of the switch, so that the number of the signal lines of the cable can be reduced.

SUMMARY OF THE INVENTION

Incidentally, in the ultrasound probe, there is a demand for securing a large opening width in order to improve an azimuth resolution of the ultrasound image. Here, the opening width is a width in an arrangement direction of the transducer array, and specifically is the number of a group of transducers capable of simultaneously transmitting and receiving the ultrasonic wave among the transducers included in the transducer array.

For example, as in JP2012-152317A, in a case where the plurality of transducers included in the transducer array are divided into half (A group and B group) and the connection destination of the signal line from the apparatus main body is switched to either the transducer in the A group or the transducer in the B group, the maximum opening width is half the number of the transducers included in the transducer array.

In addition, in order to form a suitable ultrasound image, an opening of the ultrasound probe may be moved in the arrangement direction of the transducer array. For example, JP2012-152317A discloses that the transducer (a transducer that transmits and receives the ultrasonic wave) to which the signal line from the apparatus main body is connected is moved little by little in the arrangement direction of the transducer array. In the example of JP2012-152317A, in a case where the A group has 192 transducers of #1 to #192 and the B group has 192 transducers of #193 to #384, from a state where the transducers of #1 to #192 are connected to the apparatus main body, the switches are switched such that the transducer of #193 is connected to the apparatus main body instead of #1 and then the transducer of #194 is connected to the apparatus main body instead of #2. However, in order to perform such control, it is necessary to individually control the plurality of switches. Therefore, in this case, the cable connecting the apparatus main body and the ultrasound probe needs to have as many control signal lines as the number of the plurality of switches, and the number of the signal lines of the cable may not be reduced.

An object of the ultrasound probe disclosed in the present specification is to reduce the number of the signal lines connecting the ultrasound probe and the apparatus main body of the ultrasound diagnostic apparatus while suppressing reduction of the maximum opening width of the ultrasound probe.

An ultrasound probe disclosed in the present specification comprises: a transducer array consisting of a plurality of transducers that irradiate a subject with an ultrasonic wave and receive a reflected wave from the subject; and a connection circuit for electrically connecting an apparatus main body of an ultrasound diagnostic apparatus and the transducer array, in which the transducer array includes a plurality of main transducers arranged in a major axis direction and consisting of a plurality of center-side main transducers located on a center side of the arrangement, a plurality of first end-side main transducers located on a first end side of the arrangement with respect to the plurality of center-side main transducers, and a plurality of second end-side main transducers located on a second end side opposite to the first end side of the arrangement with respect to the plurality of center-side main transducers, and the connection circuit includes a first circuit connecting the apparatus main body and the center-side main transducer, and a second circuit having a plurality of main transducer switching switches for switching a connection destination of the apparatus main body between the first end-side main transducer and the second end-side main transducer.

In addition, each of the main transducer switching switches may switch the connection destination of the apparatus main body between a k-th first end-side main transducer from the first end side among the plurality of first end-side main transducers in the arrangement and a k-th second end-side main transducer from a side of the plurality of center-side main transducers among the plurality of second end-side main transducers.

With the configuration, the connection destination (a connection destination of one signal line included in a cable connecting the apparatus main body and the ultrasound probe) of the apparatus main body is switched between the first end-side main transducer and the second end-side main transducer by the second circuit. Therefore, the number of the signal lines of the cable can be reduced by the number of the first end-side main transducers or the second end-side main transducers.

Further, with the configuration, in a state where the apparatus main body and the first end-side main transducer are connected, the opening of the ultrasound probe is moved from the first end side toward the second end side, and in a case where a second end-side end portion of the opening reaches a second end-side end portion of the center-side main transducer, the connection destination of the apparatus main body can be controlled to be switched from the first end-side main transducer to the second end-side main transducer, and then the opening can be controlled to be moved to the second end side until it reaches a second end. As a result, the plurality of main transducer switching switches need only be controlled simultaneously, so that the reduction of the opening width of the ultrasound probe can be suppressed (the opening width is maintained at a width corresponding to the number of the center-side main transducers) while reducing the number of control signal lines (that is, the number of the signal lines of the cable) required to control the main transducer switching switches.

The first circuit may be provided on a first substrate, and the second circuit may be provided on a second substrate different from the first substrate.

With the configuration, the first substrate and the second substrate can be used for various ultrasound probes having various numbers of the transducers.

The transducer array may include a plurality of sub-transducers disposed on both sides of the respective main transducers in a minor axis direction orthogonal to the major axis direction, and the first circuit and the second circuit may each include a sub-transducer switching switch for switching between connection and non-connection between a signal line connected to the main transducer and the sub-transducers disposed on both sides of the main transducer in the minor axis direction.

With the configuration, the opening width in the minor axis direction can be switched.

An ultrasound diagnostic apparatus disclosed in the present specification comprises: the ultrasound probe according to claim **1**, a transmission and reception circuit that transmits a transmission signal to the ultrasound probe, and that performs signal processing on a reception signal from the ultrasound probe; and a switch control circuit that controls switching of the main transducer switching switch, wherein, in a state in which the apparatus main body and the first end-side main transducer are connected to each other, the transmission and reception circuit transmits the transmission signal to the first end-side main transducer and the plurality of the center-side main transducers, and then moves an opening formed from a plurality of the main transducers which are targets of transmission of the transmission signal toward the second end side, the switch control circuit switches the main transducer switching switch so that the apparatus main body and the second end-side main transducer are connected to each other, when an end of the opening on the second end side has reached an end of the center-side main transducer on the second end side, and, in a state in which the apparatus main body and the second end-side main transducer are connected to each other, the transmission and reception circuit transmits the transmission signal to the plurality of the center-side main transducers and the second end-side main transducer, and then moves the opening toward the second end side.

The plurality of the first end-side main transducers may be divided into a plurality of first end-side blocks, each of which is a unit of switching control of the main transducer switching switch, and each including a plurality of the first end-side main transducers which are successive in the arrangement, the plurality of the second end-side transducers may be divided into a plurality of second end-side blocks, each of which is a unit of switching control of the main transducer switching switch, each corresponding respectively to each of the first end-side blocks according to the arrangement, and each including a plurality of the second end-side main transducers which are successive in the arrangement, in a state in which the apparatus main body and each of the first end-side blocks are connected to each other, the transmission and reception circuit may transmit the transmission signal to the first end-side block and some of the plurality of the center-side main transducers, and may then move an opening formed from a plurality of the main transducers which are targets of transmission of the transmission signal toward the second end side, the switch control circuit may switch the main transducer switching switch so that the apparatus main body is disconnected from a first end-side first block which is the first end-side block positioned at a position closest to the first end, and the apparatus main body and a second end-side first block which is the second end-side block adjacent to the center-side main transducer are connected to each other, when an end of the opening on the second end side has reached an end of the center-side main transducer on the second end side, and, in a state in which the apparatus main body is connected to the first end-side block(s) other than the first end-side first block and the second end-side first block, the transmission and reception circuit may transmit the transmission signal to the first end-side block(s) other than the first end-side first block, the plurality of the center-side main transducers, and a plurality of the second end-side main transducers included in the second end-side first block, and may then move the opening toward the second end side.

A maximum width APmax of the opening may be represented by: APmax=USch−N, wherein USch=T−(B×N), wherein T represents the number of the plurality of transducers, B represents the number of the first end-side blocks or the second end-side blocks, and N represents the number of transducers included in one first end-side block or one second end-side block.

With the ultrasound probe disclosed in the present specification, it is possible to reduce the number of the signal lines connecting the ultrasound probe and the apparatus main body of the ultrasound diagnostic apparatus while suppressing reduction of the maximum opening width of the ultrasound probe.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a block diagram of an ultrasound diagnostic apparatus according to the present embodiment.

FIG. **2** is a schematic diagram of a transducer array.

FIG. **3** is a circuit diagram of a connection circuit.

FIG. **4** is a circuit diagram of a control signal line for controlling each switch of the connection circuit in a first embodiment of the present disclosure.

FIG. **5** is a diagram showing a state in which an opening of an ultrasound probe moves in the first embodiment of the present disclosure.

FIG. **6** is a conceptual diagram showing a plurality of first end-side blocks and a plurality of second end-side blocks in a second embodiment of the present disclosure.

FIG. **7** is a circuit diagram showing a connection relationship among a first end-side block, a second end-side block, a switch provided in a connection circuit, and a control signal line for controlling the switch, in the second embodiment of the present disclosure.

FIG. **8** is a diagram showing a state in which an opening of an ultrasound probe moves in the second embodiment of the present disclosure.

FIG. **9** is a diagram showing the opening when a maximum width of the opening is set larger than USch−N.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

<First Embodiment> FIG. **1** is a block diagram of an ultrasound diagnostic apparatus **10** according to the present embodiment. The ultrasound diagnostic apparatus **10** is a medical apparatus that is installed in a medical institution such as a hospital and that is used for an ultrasound examination.

The ultrasound diagnostic apparatus **10** is an apparatus that scans a subject with an ultrasonic beam and that generates an ultrasound image based on a reception signal obtained by the scanning. For example, the ultrasound diagnostic apparatus **10** forms a tomographic image (B-mode image) in which an amplitude intensity of a reflected wave from a scanning surface is converted into brightness based on the reception signal. Alternatively, the ultrasound diagnostic apparatus **10** forms a Doppler image, which is an ultrasound image representing a motion speed of a tissue in the subject, based on a difference (Doppler shift) in frequency between a transmission wave and a reception wave.

The ultrasound diagnostic apparatus **10** is configured to include an ultrasound probe **12**, an apparatus main body **14**, a monitor **16**, a cable **18**, and a connector box **20**.

The ultrasound probe **12** is a device that transmits and receives an ultrasonic wave to and from the subject. Details of the ultrasound probe **12** will be described below.

The apparatus main body **14** is a device that performs a process of transmitting a transmission signal for causing the ultrasound probe **12** to transmit an ultrasonic wave, a process of forming an ultrasound image based on a reception signal from the ultrasound probe **12**, or the like. Although the apparatus main body **14** includes various circuits for performing various processes, only a transmission and reception circuit **30** that transmits the transmission signal to the ultrasound probe **12** and performs signal processing on the reception signal from the ultrasound probe **12**, and a switch control circuit **31** for controlling switching of a switch provided in a connection circuit **42**, to be described later, are shown in FIG. **1**.

The monitor **16** is composed of, for example, a liquid crystal display, an organic electro luminescence (EL), or the like. The ultrasound image formed by the apparatus main body **14** is displayed on the monitor **16**.

The cable **18** electrically connects the ultrasound probe **12** and the apparatus main body **14**. Specifically, the cable **18** has a plurality of signal lines, and each signal line connects each transducer included in a transducer array **40** of the ultrasound probe **12** and the transmission and reception circuit **30** of the apparatus main body **14**. The connector box **20** is connected to the apparatus main body **14** side of the cable **18**. By attaching the connector box **20** to a connector provided in the apparatus main body **14**, the ultrasound probe **12** and the apparatus main body **14** are connected via the cable **18** and the connector box **20**. In addition, as shown in FIG. **1**, a shift register **32** is provided in the connector box **20**. A function of the shift register **32** will be described below.

Hereinafter, the details of the ultrasound probe **12** will be described. The ultrasound probe **12** is configured to include the transducer array **40** and a connection circuit **42**.

The transducer array **40** is composed of a plurality of transducers that irradiate the subject with an ultrasonic wave or receive a reflected wave from the subject.

FIG. **2** is a schematic diagram of the transducer array **40**. The transducer array **40** is configured to include a plurality of main transducers **50** arranged in one direction. In the present specification, an arrangement direction of the main transducers **50** is referred to as a "major axis direction". The transducer array **40** is configured to include, for example, 320 main transducers **50**, but the number of the main transducers **50** is not limited to this.

The plurality of main transducers **50** arranged in the major axis direction are conceptually divided into a plurality of center-side main transducers **50*a*** located on a center side of the arrangement, a plurality of first end-side main transducers **50*b*** located on a first end side (upper end side in FIG. **2**) of the arrangement with respect to the plurality of center-side main transducers **50*a***, and a plurality of second end-side main transducers **50*c*** located on a second end side (lower end side in FIG. **2**) opposite to the first end side of the arrangement with respect to the plurality of center-side main transducers **50*a***. The number of the first end-side main transducers **50*b*** is the same as the number of the second end-side main transducers **50*c***. The number of the center-side main transducers **50*a*** and the number of the first end-side main transducers **50*b*** or the number of the second end-side main transducers **50*c*** may be the same as or different from each other.

In addition, in the present embodiment, the transducer array **40** is configured to include a plurality of sub-transducers **52** disposed on both sides of the respective main transducers **50** in a minor axis direction orthogonal to the major axis direction. The sub-transducer **52** exerts an effect of widening an opening in the minor axis direction. In the present embodiment, one sub-transducer **52** is provided on each of both sides of each main transducer **50** (two sub-transducers **52** for one main transducer **50**), but the plurality of sub-transducers **52** may be provided on each of both sides of each main transducer **50**.

As described above, since the plurality of main transducers **50** are arranged in the major axis direction, the plurality of sub-transducers **52** are also arranged in the major axis direction. The plurality of main transducers **50** include the center-side main transducers **50*a***, the first end-side main transducers **50*b***, and the second end-side main transducers **50*c***, and similarly, the plurality of sub-transducers **52** are also conceptually divided into a plurality of center-side sub-transducers **52*a*** located on a center side of the arrangement, a plurality of first end-side sub-transducers **52*b*** located on a first end side (upper end side in FIG. **2**) of the arrangement with respect to the plurality of center-side sub-transducers **52*a***, and a plurality of second end-side sub-transducers **52***c* located on a second end side (lower end side in FIG. **2**) opposite to the first end side of the arrangement with respect to the plurality of center-side sub-transducers **52***a*. The center-side sub-transducer **52***a* is disposed on both sides in the minor axis direction of the center-side main transducer **50***a*, the first end-side sub-transducer **52***b* is disposed on both sides in the minor axis direction of the first end-side main transducer **50***b*, and the second end-side sub-transducer **52***c* is disposed on both sides in the minor axis direction of the second end-side main transducer **50***c*.

In the present specification, the sub-transducers **52** disposed on both sides of the main transducer **50** in the minor axis direction may be described as the sub-transducers **52** corresponding to the (relevant) main transducer **50**.

The connection circuit **42** is a circuit provided between the transmission and reception circuit **30** and the transducer array **40** in a signal path of a transmission signal from the transmission and reception circuit **30** (that is, the apparatus main body **14**) and a reception signal from the transducer array **40**, and is a circuit for electrically connecting the transmission and reception circuit **30** and the transducer array **40**. Specifically, the connection circuit **42** is a circuit that electrically connects each transducer (the main transducer **50** and the sub-transducer **52**) included in the transducer array **40** and the cable **18** (more specifically, each signal line included in the cable **18**) connected to the transmission and reception circuit **30**.

FIG. **3** is a circuit diagram of the connection circuit **42**. Hereinafter, the connection circuit **42** will be described with reference to FIG. **3**. In FIG. **3**, on the right side of the connection circuit **42**, the main transducer **50** and two sub-transducers **52** corresponding to the main transducer **50** are represented by one symbol M. A plurality of the symbols M arranged vertically represent the main transducers **50** and the sub-transducers **52** arranged in the major axis direction. A numerical value (#) described on the right side of each symbol M represents an order of the main transducer **50** or the sub-transducer **52** from the first end side in the arrangement. In the example of FIG. **3**, 320 main transducers **50** are arranged in the major axis direction. In addition, in the example of FIG. **3**, the number of the center-side main transducers **50***a* is 192 (#65 to #256), the number of the first end-side main transducers **50***b* is 64 (#1 to #64), and the number of the second end-side main transducers **50***c* is 64 (#257 to #320).

The connection circuit **42** is configured to include a first circuit **60** and a second circuit **62**. The first circuit **60** is a circuit that connects the transmission and reception circuit **30** and the center-side main transducer **50***a*, and the second circuit **62** is a circuit having a function of switching a connection destination of the transmission and reception circuit **30** between the first end-side main transducer **50***b* and the second end-side main transducer **50***c*. Hereinafter, in the present specification, each signal path included in the first circuit **60** or the second circuit **62** will be referred to as a "pattern".

The first circuit **60** includes a main pattern **60***a* that connects the transmission and reception circuit **30** and the center-side main transducer **50***a*. Specifically, the main pattern **60***a* connects a transmission signal line S of the cable **18** and the center-side main transducer **50***a*. Therefore, as long as the connector box **20** is connected to the apparatus main body **14**, the center-side main transducer **50***a* is always connected to the transmission and reception circuit **30** by the main pattern **60***a*.

In addition, the first circuit **60** includes a sub-pattern **60***b* that is branched from the main pattern **60***a* and is connected to the center-side sub-transducers **52***a* corresponding to the center-side main transducer **50***a* to which the main pattern **60***a* is connected. A switch SW**1** as a sub-transducer switching switch is provided in the middle of the sub-pattern **60***b*. The switch SW**1** is a switch for switching connection/non-connection between the main pattern **60***a* and the center-side sub-transducer **52***a*. In a case where the switch SW**1** is ON, the center-side main transducer **50***a* and the center-side sub-transducers **52***a* corresponding to the center-side main transducer **50***a* are connected to the same main pattern **60***a* (that is, the same transmission signal line S). In a case where the switch SW**1** is OFF, the center-side sub-transducer **52***a* is not connected to the main pattern **60***a* (that is, the transmission signal line S).

The first circuit **60** is provided for each center-side main transducer **50***a* (or center-side sub-transducers **52***a* corresponding to the center-side main transducer **50***a*).

The second circuit **62** includes a cable-side pattern **62***a* connected to the transmission signal line S of the cable **18**, a first end-side main pattern **62***b* connected to the first end-side main transducer **50***b*, and a second end-side main pattern **62***c* connected to the second end-side main transducer **50***c*. A switch SW**2** as a main transducer switching switch is provided between the cable-side pattern **62***a*, and the first end-side main pattern **62***b* and the second end-side main pattern **62***c*. The switch SW**2** is a switch for switching a connection destination of the cable-side pattern **62***a* to either the first end-side main pattern **62***b* or the second end-side main pattern **62***c*. In other words, the connection destination of the transmission and reception circuit **30** is switched between the first end-side main transducer **50***b* and the second end-side main transducer **50***c* by the switch SW**2**.

Although the second circuit **62** is provided for each combination of one first end-side main transducer **50***b* and one second end-side main transducer **50***c*, the switch SW**2** included in the second circuit **62** switches the connection destination of the transmission and reception circuit **30** between a k-th first end-side main transducer **50***b* from the first end side of the arrangement of the main transducers **50** and a k-th second end-side main transducer **50***c* from the center-side main transducer **50***a* side of the arrangement. For example, the switch SW**2** of the 1st second circuit **62** switches the connection destination of the transmission and reception circuit **30** between the first end-side main transducer **50***b* which is the first (#1) in the arrangement of the plurality of main transducers **50** and the second end-side main transducer **50***c* which is the 257th (closest to the center-side main transducer **50***a* side (which may be referred to as the first end side of the arrangement) among the plurality of second end-side main transducers **50***c*) in the arrangement. In addition, the switch SW**2** of the 2nd second circuit **62** switches the connection destination of the transmission and reception circuit **30** between the first end-side main transducer **50***b* which is the second (#2) in the arrangement of the plurality of main transducers **50** and the second end-side main transducer **50***c* which is the 258th (the second from the center-side main transducer **50***a* side (the first end side of the arrangement) among the plurality of second end-side main transducers **50***c*) in the arrangement.

In addition, the second circuit **62** includes a first end-side sub-pattern **62***d* connected to the first end-side sub-transducer **52***b* and a second end-side sub-pattern **62***e* connected to the second end-side sub-transducer **52***c*. A switch SW**3** as a sub-transducer switching switch is provided between the cable-side pattern **62***a*, and the first end-side sub-pattern **62***d* and the second end-side sub-pattern **62***e*. The switch SW**3** is a switch for switching a connection destination of the cable-side pattern **62***a* to either the first end-side sub-pattern **62***d* or the second end-side sub-pattern **62***e*. In other words, the connection destination of the transmission and reception circuit **30** is switched between the first end-side sub-transducer **52***b* and the second end-side sub-transducer **52***c* by the switch SW**3**.

As with the switch SW**2**, the switch SW**3** included in one second circuit **62** switches the connection destination of the transmission and reception circuit **30** between a k-th first end-side sub-transducer **52***b* from the first end side of the arrangement of the sub-transducers **52** and a k-th second end-side sub-transducer **52***c* from the center-side sub-transducer **52***a* side of the arrangement. For example, the switch SW**3** of the 1st second circuit **62** switches the connection destination of the transmission and reception circuit **30** between the first end-side sub-transducer **52***b* which is the first (#1) in the arrangement of the plurality of sub-transducers **52** and the second end-side sub-transducer **52***c* which is the 257th (closest to the center-side sub-transducer **52***a* side (which may be referred to as the first end side of the arrangement) among the plurality of second end-side sub-transducers **52***c*) in the arrangement. In addition, the switch SW**3** of the 2nd second circuit **62** switches the connection destination of the transmission and reception circuit **30** between the first end-side sub-transducer **52***b* which is the second (#2) in the arrangement of the plurality of sub-transducers **52** and the second end-side sub-transducer **52***c* which is the 258th (the second from the center-side sub-transducer **52***a* side (the first end side of the arrangement) among the plurality of second end-side sub-transducers **52***c*) in the arrangement.

In the present embodiment, the connection circuit **42** is divided and provided on a plurality of substrates. Specifically, the first circuit **60** is provided on a first substrate **70**, and the second circuit **62** is provided on a second substrate **72** different from the first substrate **70**. In the present embodiment, the second circuit **62** is not provided on the first substrate **70**, and the first circuit **60** is not provided on the second substrate **72**.

For example, in the example of FIG. **3**, the number of the provided center-side main transducers **50***a* is 192, so that 192 first circuits **60** are provided on the first substrate **70**. A plurality of the first circuits **60** may be divided and provided on a plurality of the first substrates **70**. For example, 96 first circuits **60** may be provided on the 1st first substrate **70**, and 96 first circuits **60** may be provided on the 2nd first substrate **70**. In this case, the ultrasound probe **12** is provided with two first substrates **70**. The number of the first circuits **60** provided on one first substrate **70** may be appropriately determined.

In addition, in the example of FIG. **3**, the number of the provided combinations of the first end-side main transducer **50***b* and the second end-side main transducer **50***c* are 64, so that 64 second circuits **62** are provided on the second substrate **72**. A plurality of the second circuits **62** may be divided and provided on a plurality of the second substrates **72**. For example, 32 second circuits **62** may be provided on the 1st second substrate **72**, and 32 second circuits **62** may be provided on the 2nd second substrate **72**. In this case, the ultrasound probe **12** is provided with two second substrates **72**. The number of the second circuits **62** provided on one second substrate **72** may also be appropriately determined.

FIG. **4** is a circuit diagram of a control signal line CS for controlling each switch (SW**1**, SW**2**, SW**3**) of the connection circuit **42**. A control signal for controlling each switch is output from a switch control circuit **31**. Note that a control signal of a dynamic signal is output to the shift register **32** from the switch control circuit **31**, and the control signal is converted into a static signal by the shift register **32**.

From the shift register **32**, two control signal lines CS**1** and CS**2** are connected to the connection circuit **42** via the cable **18**.

The control signal line CS**1** is connected in parallel to a plurality of the switches SW**1** included in the plurality of first circuits **60** provided on the first substrate **70** and a plurality of the switches SW**3** included in the plurality of second circuits **62** provided on the second substrate **72**. Accordingly, the control signal from the shift register **32** to the switches SW**1** is supplied in parallel to the plurality of switches SW**1** and the plurality of switches SW**3**. The switch SW**1** is turned on in a case where a value of the control signal line CS**1** is high (H), and the switch SW**1** is turned off in a case where the value of the control signal line CS**1** is low (L). The switch SW**3** is enabled in a case where the value of the control signal line CS**1** is H, and the switch SW**3** is turned off in a case where the value of the control signal line CS**1** is L. The state where the switch SW**3** is enabled means a state where the cable-side pattern **62***a* is connected to either the first end-side sub-pattern **62***d* or the second end-side sub-pattern **62***e*. The state where the switch SW**3** is turned off means a state where the cable-side pattern **62***a* is connected to neither the first end-side sub-pattern **62***d* nor the second end-side sub-pattern **62***e*.

In the present embodiment, the control signal line CS**2** is connected in parallel to the plurality of switches SW**2** and the plurality of switches SW**3** included in the plurality of second circuits **62** provided on the second substrate **72**. Accordingly, the control signal from the shift register **32** to the switches SW**2** is supplied in parallel to the plurality of switches SW**2**. In a case where a value of the control signal line CS**2** is L, the switch SW**2** is in a state where the cable-side pattern **62***a* and the first end-side main pattern **62***b* are connected, and in a case where the value of the control signal line CS**2** is H, the switch SW**2** is in a state where the cable-side pattern **62***a* and the second end-side main pattern **62***c* are connected. That is, the plurality of switches SW**2** included in the plurality of second circuits **62** are simultaneously controlled.

In the present embodiment, the control signal line CS**2** is connected in parallel to the plurality of switches SW**3**. Accordingly, the control signal from the shift register **32** to the switches SW**3** is supplied in parallel to the plurality of switches SW**3**. In a case where the value of the control signal line CS**1** is H and the value of the control signal line CS**2** is L, the switch SW**3** is in a state where the cable-side pattern **62***a* and the first end-side sub-pattern **62***d* are connected, and in a case where the value of the control signal line CS**1** is H and the value of the control signal line CS**2** is H, the switch SW**3** is in a state where the cable-side pattern **62***a* and the second end-side sub-pattern **62***e* are connected. That is, the plurality of switches SW**3** included in the plurality of second circuits **62** are simultaneously controlled.

The outline of the ultrasound probe **12** according to the present embodiment is as described above. In the ultrasound probe **12**, one transmission signal line S of the cable **18** corresponds to two main transducers **50** (the first end-side main transducer **50***b* and the second end-side main transducer **50***c*) by the second circuit **62**, so that the number of the transmission signal lines S can be reduced by the number of the first end-side main transducers **50***b* (or the second end-side main transducers **50***c*).

In addition, in the arrangement direction of the main transducers **50**, the connection destination of the transmission and reception circuit **30** is switched between the first end-side main transducers **50***b* and the second end-side main transducers **50***c* provided on both sides of the center-side main transducers **50***a*, whereby the reduction of the maximum opening width in the arrangement direction (major axis direction) of the main transducers **50** is suppressed. This will be described with reference to FIG. **5**.

In the example of FIG. **5**, an example is shown in which an opening A (corresponding to 160 transducers in the example of FIG. **5**) with the maximum opening width is moved in a direction from the first end side to the second end side of the arrangement of the main transducers **50**, that is, in a direction from the first (#1) to the 320th (#320) in the arrangement of the main transducers **50**. Although FIG. **5** also shows the sub-transducer **52**, the following description of the movement of the opening A in the major axis direction will be made with attention to the main transducer **50**.

The figure on the left side of FIG. **5** shows a state where the opening A is the main transducers **50** of #1 to #160. In this case, the switch control circuit **31** controls the control signal line CS**2** to L to connect the transmission and reception circuit **30** and the first end-side main transducers **50***b* (#1 to #64). In addition, the transmission and reception circuit **30** transmits a transmission signal to all of the first end-side main transducers **50***b* (#1 to #64) and a part of the center-side main transducers **50***a* (#65 to #160).

From this state, the transmission and reception circuit **30** gradually moves the opening A to the second end side (the side of the main transducer **50** of #320) of the arrangement of the main transducers **50** while maintaining the width of the opening A, while the switch control circuit **31** maintains the control signal line CS**2** at L. That is, the transmission and reception circuit **30** gradually changes the main transducer **50** of a transmission destination of the transmission signal to the side of the main transducer **50** of #320. Eventually, an end of the opening A on the second end side reaches an end of the center-side main transducers **50***a* on the second end side, and the opening A becomes the main transducers **50** of #97 to #256 as shown in the middle figure of FIG. **5**.

In this state, the switch control circuit **31** changes the control signal line CS**2** to H and switches the connection destination of the transmission and reception circuit **30** from the first end-side main transducers **50***b* (#1 to #64) to the second end-side main transducers **50***c* (#257 to #320). Then, the transmission and reception circuit **30** transmits the transmission signal to some of the center-side main transducers **50***a* (#98 to #256) and the first of the second end-side main transducers **50***c* (#257).

Hereinafter, the transmission and reception circuit **30** gradually moves the opening A to the second end side (the side of the main transducer **50** #320) of the arrangement of the main transducers **50** while maintaining the width of the opening A, while the switch control circuit **31** maintains the control signal line CS**2** at H. That is, the transmission and reception circuit **30** gradually changes the main transducer **50** of a transmission destination of the transmission signal to the side of the main transducer **50** #320. Eventually, the opening A becomes the main transducers **50** #161 to #320 as shown in the right figure in FIG. **5**.

As described above, in the present embodiment, in the arrangement direction of the main transducers **50**, the connection destination of the transmission and reception circuit **30** is switched between the first end-side main transducers **50***b* and the second end-side main transducers **50***c* provided on both sides of the center-side main transducers **50***a*, whereby reduction of the maximum width of the opening A can be suppressed. In addition, the plurality of switches SW**2** for switching the connection destination of the transmission and reception circuit **30** between the first end-side main transducers **50***b* and the second end-side main transducers **50***c* need only be simultaneously controlled, so that the number of the control signal lines CS**2** for switching the switches SW**2** may be one. That is, according to the present embodiment, it is possible to reduce the number of the signal lines of the cable **18** while suppressing the reduction of the maximum opening width of the ultrasound probe **12**.

In addition, in the present embodiment, the plurality of first circuits **60** are provided on the first substrate **70**, and the plurality of second circuits **62** are provided on the second substrate **72**. Accordingly, the first substrate **70** and the second substrate **72** can be used for various ultrasound probes **12** having various numbers of the transducers. For example, in a case where the first substrate **70** provided with 96 first circuits **60** and the second substrate **72** provided with 32 second circuits **62** are prepared, the ultrasound probe **12** having 320 transducers need only be provided with two first substrates **70** and two second substrates **72**, the ultrasound probe **12** having 256 transducers need only be provided with two first substrates **70** and one second substrate **72**, and the ultrasound probe **12** having 160 transducers need only be provided with one first substrate **70** and one second substrate **72**.

In addition, in the present embodiment, by controlling the switch SW**1**, it is possible to also connect, to the transmission and reception circuit **30**, the sub-transducers **52** corresponding to the main transducers **50** connected to the transmission and reception circuit **30**. By switching the connection/non-connection of the transmission and reception circuit **30** to the sub-transducers **52**, it becomes possible to vary an area for transmission and reception of the ultrasound, and to consequently adjust focus of the ultrasonic beam.

<Second Embodiment> A block diagram of an ultrasound diagnostic apparatus **10** according to a second embodiment of the present disclosure, a schematic diagram of the transducer array **40**, and a circuit diagram of the connection circuit **42** are in the most part similar to those of the first embodiment of the present disclosure. The ultrasound diagnostic apparatus **10** according to the second embodiment will now be described primarily with regard to portions that are different from the first embodiment.

FIG. **6** is a conceptual diagram showing a plurality of first end-side blocks **80** and a plurality of second end-side blocks **82** defined in the transducer array **40**. In the second embodiment, the plurality of the first end-side main transducers **50***b* are conceptually divided into the plurality of first end-side blocks **80**. Each first end-side block **80** also includes a plurality of the first end-side sub-transducers **52***b*, but in the following, description will be given with attention to the main transducers **50**, and the sub-transducers **52** will be described later. Each first end-side block **80** is formed including a plurality of first end-side transducers **50***b* which are successive in the arrangement of the transducer array **40**. In the present embodiment, as shown in FIG. **6**, the first end-side blocks **80** are divided into two first end-side blocks **80**, including a first end-side first block **80***a* including the first end-side main transducers **50***b* of #1 to #32, and a first end-side second block **80***b* including the first end-side main transducers **50***b* of #33 to #64. In this manner, in the present embodiment, each first end-side block **80** includes 32 first end-side main transducers **50***b*. The number of the first end-side blocks **80** is not limited to two, and may be another number. As will be described later in detail, the first end-side block **80** is a unit for performing switching control of the switch SW**2** serving as the main transducer switching switch.

Similarly, in the second embodiment, the plurality of the second end-side main transducers **50***c* are conceptually divided into a plurality of second end-side blocks **82**. Similar to the first end-side block **80**, each second end-side block **82** also includes a plurality of the second end-side sub-transducers **52***c*, but in the following, description will be given with attention to the main transducers **50**, and the sub-transducers **52** will be described later. Each second end-side block **82** is also formed including a plurality of the second end-side main transducers **50***c* which are successive in the arrangement of the transducer array **40**. In the present embodiment, as shown in FIG. **6**, the second end-side blocks **82** are divided into two second end-side blocks **82** including a second end-side first block **82***a* including the second end-side main transducers **50***c* #257 to #288, and a second end-side second block **82***b* including the second end-side main transducers **50***c* #289 to #320. In this manner, in the present embodiment, each second end-side block **82** includes 32 second end-side main transducers **50***c*. In particular, each second end-side block **82** is provided corresponding to each first end-side block **80**, according to the arrangement of the transducer array **40**. In the present embodiment, a second end-side first block **82***a* positioned at the closest position to the first end among the second end-side blocks **82** (that is, adjacent to the center-side main transducer **50***a*) corresponds to the first end-side first block **80***a* which is positioned at the closest position to the first end among the first end-side blocks **80**. In addition, a second end-side second block **82***b* which is at the second position from the first end side among the second end-side blocks **82** corresponds to a first end-side second block **80***b* at the second position from the first end side among the first end-side blocks **80**. In this manner, the number of the first end-side blocks **80** and the number of the second end-side blocks **82** are equal to each other. In addition, the number of the first end-side main transducers **50***b* included in the first end-side block **80** and the number of the second end-side main transduces **50***c* included in the second end-side block **82** corresponding to the first end-side block **80** are equal to each other. The second end-side block **82** is also a unit for performing the switching control of the switch SW**2**.

FIG. **7** is a circuit diagram showing a connection relationship among the first end-side block **80**, the second end-side block **82**, the switches SW**2** and SW**3**, the control signal line CS**1**, and the control signal line CS**2** for controlling the switches SW**2** and SW**3**. In the second embodiment also, a switch SW**1** (refer to FIG. **4**), to which the plurality of center-side main transducers **50***a* (and the plurality of center-side sub-transducers **52***a*) and the control signal line CS**1** are connected, is provided, but illustration thereof is omitted in FIG. **7**.

Similar to the first embodiment, the switch SW**2** changes the connection destination of the transmission and reception circuit **30** (refer to FIG. **1**) between the first end-side main transducer **50***b* and the second end-side main transducer **50***c*. In particular, in the second embodiment, the switch SW**2** switches the connection destination of the transmission and reception circuit **30** between a first end-side main transducer **50***b* included in the first end-side block **80**, and a second end-side main transducer **50***c* included in the second end-side block **82** corresponding to the first end-side block **80**. In the example configuration of FIG. **7**, a switch SW**2-1** switches between a first end-side main transducer **50***b* included in the first end-side first block **80***a*, and a second end-side main transducer **50***c* included in the second end-side first block **82***a*, and a switch SW**2-2** switches between a first end-side main transducer **50***b* included in the first end-side second block **80***b* and a second end-side main transducer **50***c* included in the second end-side second block **82***b*. While the example configuration of FIG. **7** only shows one switch SW**2** for one set of the first end-side block **80** and the second end-side block **82** (for example, only one switch SW**2-1** is shown for the set of the first end-side first block **80***a* and the second end-side first block **82***a*), in reality, the switches SW**2** for the set are provided in a number corresponding to the number of the main transducers **50** included in the first end-side block **80** and the second end-side block **82**. In the present specification, a plurality of switches SW**2** for the set of the first end-side first block **80***a* and the second end-side first block **82***b* will be collectively referred to as the "switch SW**2-1**", and a plurality of switches SW**2** for the set of the first end-side second block **80***b* and the second end-side second block **82***b* will be collectively referred to as the "switch SW**2-2**".

In the first embodiment, one control signal line CS**2** is connected in parallel to a plurality of switches SW**2** and a plurality of switches SW**3**. That is, switching of all of the switches SW**2** and all of the switches SW**3** are simultaneously controlled. On the other hand, in the second embodiment, separate control signal lines CS**2** are provided respectively for the plurality of switches SW**2** corresponding to a plurality of sets of the first end-side blocks **80** and the second end-side blocks **82**. In the example configuration of FIG. **7**, a control signal line CS**2-1** for transmitting the control signal from the shift register **32** to the switch SW**2-1** for the set of the first end-side first block **80***a* and the second end-side first block **82***a* is provided, and a control signal line CS**2-2** for transmitting the control signal from the shift register **32** to the switch SW**2-2** for the set of the first end-side second block **80***b* and the second end-side second block **82***b* is provided. With this configuration, in the second embodiment, the switches SW**2** corresponding to the respective sets of the first end-side blocks **80** and the second end-side blocks **82** can be individually controlled. That is, in the second embodiment, the connection destination of the transmission and reception circuit **30** can be switched individually for each set of the first end-side block **80** and the second end-side block **82**.

In the present embodiment, when the value of the control signal line CS**2-1** is L, the switch SW**2-1** is set to a state in which the transmission and reception circuit **30***a* and the first end-side main transducers **50***b* included in the first end-side first block **80***a* are connected to each other, and, when the value of the control signal line CS**2-1** is H, the switch SW**2-1** is set to a state in which the transmission and reception circuit **30***a* and the second end-side main transducers **50***c* included in the second end-side first block **82***a* are connected to each other. Similarly, in the present embodiment, when the value of the control signal line CS**2-2** is L, the switch SW**2-2** is set to a state in which the transmission and reception circuit **30***a* and the first end-side main transducers **50***b* included in the first end-side second block **80***b* are connected to each other, and, when the value of the control signal line CS**2-2** is H, the switch SW**2-2** is set to a state in which the transmission and reception circuit **30***a* and the second end-side main transducers **50***c* included in the second end-side second block **82***b* are connected to each other.

The switching control of the switch SW**2** and the transmission control of the transmission signal for moving the opening A in the major axis direction in the second embodiment will now be described with reference to FIG. **8**. FIG.

8 also shows the sub-transducers **52**, but the description will be given with attention to the main transducers **50**.

The figure on the left side of FIG. **8** shows a state where the opening A is the main transducers **50** #1 to #224. As will be described later in detail, when the number of the main transducers **50** is 320, the number of the first end-side blocks **80** (or the second end-side blocks **82**) is 2, and the number of the main transducers **50** included in each first end-side block **80** (or each second end-side block **82**) is 32, the maximum opening width is a width corresponding to 224 main transducers **50**, as shown in FIG. **8**.

When the opening A is the main transducers **50** #1 to #224, the switch control circuit **31** controls the control signal lines CS**2**-**1** and CS**2**-**2** to L to connect the transmission and reception circuit **30** and the first end-side main transducers **50***b* (#1 to #64) included in first end-side blocks **80** (that is, the first end-side first block **80***a* and the first end-side second block **80***b*). In this state, the transmission and reception circuit **30** transmits the transmission signal to all of the plurality of the first end-side main transducers **50***b* (#1 to #64) included in the first end-side blocks **80** and some of the plurality of the center-side main transducers **50***a* (#65 to #224).

From this state, the transmission and reception circuit **30** gradually moves the opening A to the second end side (the side of the main transducer **50** of #320) of the arrangement of the main transducers **50** while maintaining the width of the opening A, while the switch control circuit **31** maintains the control signal lines CS**2**-**1** and CS**2**-**2** at L. That is, the transmission and reception circuit **30** gradually changes the main transducer **50** of a transmission destination of the transmission signal to the side of the main transducer **50** #320. Eventually, an end of the opening A on the second end side reaches an end of the center-side main transducers **50***a* on the second end side, and the opening A becomes the main transducers **50** #33 to #256 as shown in the middle figure of FIG. **8**.

In this state, the switch control circuit **31** changes the control signal line CS**2**-**1** to H while maintaining the control signal line CS**2**-**2** at L. With this process, the transmission and reception circuit **30** is disconnected from the first end-side main transducers **50***b* (#1 to #32) included in the first end-side first block **80***a*, and instead, the transmission and reception circuit **30** and the second end-side main transducers **50***c* (#257 to #288) included in the second end-side first block **82***a* are connected to each other. In this state, the connection destination of the transmission and reception circuit **30** is the first end-side main transducers **50***b* (#33 to #64) included in the first end-side second block **80***b*, the center-side main transducers **50***a* (#65 to #256), and the second end-side main transducers **50***c* (#257 to #288) included in the second end-side first block **82***a*.

In this state, the transmission and reception circuit **30** transmits the transmission signal to some of the first end-side main transducers **50***b* (#34 to #64) included in the first end-side block(s) **80** other than the first end-side first block **80***a* (in the present example configuration, the first end-side second block **80***b*), all of the plurality of the center-side main transducers **50***a*, and the first of the second end-side main transducers **50***c* (#257) included in the second end-side first block **82***a*.

Then, the transmission and reception circuit **30** gradually moves the opening A to the second end side (the side of the main transducer **50** #320) of the arrangement of the main transducers **50** while maintaining the width of the opening A, while the switch control circuit **31** maintains the control signal line CS**2**-**1** at H and the control signal line CS**2**-**2** at L. Eventually, an end of the opening A on the second end side reaches an end of the second end-side first block **82***a* on the second end side, and the opening A becomes the main transducers **50** #65 to #288 as shown in the right figure of FIG. **8**.

In this state, the switch control circuit **31** changes the control signal line CS**2**-**2** to L while maintaining the control signal line CS**2**-**1** at H. With this process, the transmission and reception circuit **30** is disconnected from the first end-side main transducers **50***b* (#33 to #64) included in the first end-side second block **80***b*, and instead, the transmission and reception circuit **30** and the second end-side main transducers **50***c* (#289 to #320) included in the second end-side second block **82***b* are connected to each other. In this state, the connection destination of the transmission and reception circuit **30** is the center-side main transducers **50***a* (#65 to #256), and the second end-side main transducers **50***c* (#257 to #320) included in the second end-side blocks **82** (that is, the second end-side first block **82***a* and the second end-side second block **82***b*).

In this state, the transmission and reception circuit **30** transmits the transmission signal to some of the plurality of the center-side main transducers **50***a* (#66 to #256), all of the second end-side main transducers **50***c* included in the second end-side first block **82***a* (#257 to #288), and the first of the second end-side main transducers **50***c* (#289) included in the second end-side second block **82***b*.

Then, the transmission and reception circuit **30** gradually moves the opening A to the second end side (the side of the main transducer **50** #320) of the arrangement of the main transducers **50** while maintaining the width of the opening A, while the switch control circuit **31** maintains the control signal lines CS**2**-**1** and CS**2**-**2** at H. Eventually, an end of the opening A on the second end side reaches an end of the second end-side second block **82***b* on the second end side (in the present embodiment, an end of the transducer array **40** on the second end side), and the opening A becomes the main transducers **50** #97 to #320.

As described, in the second embodiment, the plurality of the first end-side main transducers **50***b* are divided into a plurality of the first end-side blocks **80**, the plurality of the second end-side main transducers **50***c* are divided into a plurality of the second end-side blocks **82**, and, for each set of the first end-side block **80** and the second end-side block **82**, the opening A is moved while the connection destination of the transmission and reception circuit **30** is sequentially switched. In the embodiment described above, a configuration is described in which the numbers of the first end-side blocks **80** and the second end-side blocks **82** are 2, but the case in which the numbers of the first end-side blocks **80** and the second end-side blocks **82** are three or more can be similarly handled, and, for each set of the first end-side block **80** and the second end-side block **82**, the opening A is moved while the connection destination of the transmission and reception circuit **30** is sequentially switched.

In the second embodiment, with the above-described process, the opening width of the opening A may be widened at least in comparison to the first embodiment. Specifically, a maximum width APmax of the opening A is represented by the following Equation 1.

$$APmax = USch - N \quad \text{(Equation 1)}$$

In Equation 1, USch (which is referred to as an "apparatus channel" in the present specification) indicates the number of a plurality of main transducers **50** which can be connected simultaneously to the transmission and reception circuit **30**, and is represented by the following Equation 2.

$$USch = T - (B \times N) \quad \text{(Equation 2)}$$

In Equation 2, T represents the number of the plurality of main transducers **50** included in the transducer array **40**, B represents the number of the first end-side blocks **80** or the second end-side blocks **82**, and N represents the number of the main transducers **50** included in one first end-side block **80** or one second end-side block **82**.

In the present embodiment, T=320, B=2, and N=32, and USch is thus **256**. Therefore, the maximum width APmax of the opening A in the present embodiment is 256−32=224.

If the maximum width APmax of the opening A is set larger than USch−N, as shown in FIG. **9**, when the end of the opening A on the second end side enters the second end-side first block **82***a*, an end of the opening A on the first end side would remain in the first end-side first block **80***a*. As described above, the first end-side first block **80***a* and the second end-side first block **82***a* are exclusively connected to the transmission and reception circuit **30**. Thus, in this case, it becomes impossible to transmit the transmission signal to either the first end-side main transducer **50***b* included in the first end-side first block **80***a* or the second end-side main transducer **50***c* included in the second end-side first block **82***a*. In the example configuration of FIG. **9**, the transmission signal cannot be transmitted to #32 or #257 either. In order to avoid such a situation, in the present embodiment, a value obtained by subtracting the number of the main transducers **50** in one block from the apparatus channel USch is set as the maximum width APmax of the opening A.

Further, in the second embodiment, a number Cn of the signal lines of the cable **18** is USch (number of transmission signal lines S)+B (number of the control signal lines CS**2**)+1 (the control signal line CS**1**). If the transducer array **40** does not include the sub-transducers **52**, the number Cn of the signal lines of the cable **18** is USch (number of the transmission signal lines S)+B (number of the control signal lines CS**2**).

Next, the sub-transducers **52** in the second embodiment will be described. As shown in FIG. **6**, each first end-side block **80** includes the first end-side sub-transducers **52***b* corresponding to the first end-side main transducers **50***b* included therein, and each second end-side block **82** includes the second end-side sub-transducers **52***c* corresponding to the second end-side main transducers **50***c* included therein. That is, conceptually, the plurality of the first end-side sub-transducers **52***b* are also divided into the plurality of the first end-side blocks **80**, and, conceptually, the plurality of the second end-side sub-transducers **52***c* are also divided into the plurality of the second end-side blocks **82**.

As shown in FIG. **7**, in the second embodiment, the switch SW**3** switches the connection destination of the transmission and reception circuit **30** between the first end-side sub-transducers **52***b* included in the first end-side block **80**, and the second end-side sub-transducers **52***c* included in the second end-side block **82** corresponding to the first end-side block **80**. In the example configuration of FIG. **7**, a switch SW**3-1** switches between the first end-side sub-transducers **52***b* included in the first end-side first block **80***a* and the second end-side sub-transducers **52***c* included in the second end-side first block **82***a*, and a switch SW**3-2** switches between the first end-side sub-transducers **52***b* included in the first end-side second block **80***b* and the second end-side sub-transducers **52***c* included in the second end-side second block **82***b*. While the example configuration of FIG. **7** only shows one switch SW**3** for one set of the first end-side block **80** and the second end-side block **82** (for example, only one switch SW**3-1** is shown for the set of the first end-side first block **80***a* and the second end-side first block **82***a*), in reality, the switches SW**3** for the set are provided in a number corresponding to the number of pairs of the sub-transducers **52** included in the first end-side blocks **80** and the second end-side blocks **82**. In the present specification, a plurality of switches SW**3** for the set of the first end-side first block **80***a* and the second end-side first block **82***a* will be collectively referred to as the "switch SW**3-1**", and a plurality of switches SW**3** for the set of the first end-side second block **80***b* and the second end-side second block **82***b* will be collectively referred to as the "switch SW**3-2**".

In the second embodiment also, similar to the first embodiment, the control signal line CS**1** is connected in parallel to the plurality of the switches SW**3**. When the value of the control signal line CS**1** is H, the switch SW**3** is enabled, and when the value of the control signal line CS**1** is L, the switch SW**3** is turned off.

In such a structure, the focus of the ultrasonic beam can be adjusted by suitably controlling the control signal line CS**1**, similar to the first embodiment.

For example, in the state shown in the figure on the left side of FIG. **8**, that is, in a state in which the switch control circuit **31** controls the control signal lines CS**2-1** and CS**2-2** to L, and the transmission and reception circuit **30** is connected to the first end-side main transducers **50***b* (#1 to #64) included in the first end-side block **80**, and the center-side main transducers **50***a* (#65 to #256), when the switch control circuit **31** changes the control signal line CS**1** from L to H, the switch SW**3-1**, the switch SW**3-2**, and the switch SW**1** (refer to FIG. **4**) are turned on, and the first end-side sub-transducers **52***b* included in the first end-side block **80** and the center-side sub-transducers **52***a* are connected to the transmission and reception circuit **30**. In this state, the transmission and reception circuit **30** can transmit the transmission signal to the first end-side sub-transducers **52***b* included in the first end-side block **80** and the center-side sub-transducers **52***a*, and can receive reception signals from the first end-side sub-transducers **52***b* included in the first end-side block **80** and the center-side sub-transducers **52***a*.

In the state in which the switch control circuit **31** controls the control signal line CS**2-1** to H and the control signal line CS**2-2** to L, and the transmission and reception circuit **30** is connected to the first end-side main transducers **50***b* (#33 to #64) included in the first end-side second block **80***b*, the center-side main transducers **50***a* (#65 to #256), and the second end-side main transducers **50***c* (#257 to #288) included in the second end-side first block **82***a*, when the switch control circuit **31** changes the control signal line CS**1** from L to H, the switch SW**1**, the switch SW**3-1**, and the switch SW**3-2** are turned on, and the first end-side sub-transducers **52***b* included in the first end-side second block **80***b*, the center-side sub-transducers **52***a*, and the second end-side sub-transducers **52***c* included in the second end-side first block **82***a* are connected to the transmission and reception circuit **30**. In this state, the transmission and reception circuit **30** can transmit the transmission signal to the first end-side sub-transducers **52***b* included in the first end-side second block **80***b*, the center-side sub-transducers **52***a*, and the second end-side sub-transducers **52***c* included in the second end-side first block **82***a*, and can receive reception signals from the first end-side sub-transducers **52***b* included in the first end-side second block **80***b*, the center-side sub-transducers **52***a*, and the second end-side sub-transducers **52***c* included in the second end-side first block **82***a*.

Further, in the state in which the switch control circuit **31** controls the control signal lines CS**2-1** and CS**2-2** to H, and the transmission and reception circuit **30** is connected to the center-side main transducers **50**a (#65 to #256), and the second end-side main transducers **50**c (#257 to #320) included in the second end-side block **82**, when the switch control circuit **31** changes the control signal line CS**1** from L to H, the switch SW**1**, the switch SW**3-1**, and the switch SW**3-2** are turned on, and the center-side sub-transducers **52**a and the second end-side sub-transducers **52**c included in the second end-side block **82** are connected to the transmission and reception circuit **30**. In this state, the transmission and reception circuit **30** can transmit the transmission signal to the center-side sub-transducers **52**a, and the second end-side sub-transducers **52**c included in the second end-side block **82**, and can receive reception signals from the center-side sub-transducers **52**a and the second end-side sub-transducers **52**c included in the second end-side block **82**.

Although the embodiment according to the present invention has been described above, the present invention is not limited to the above-described embodiment, and various changes can be made without departing from the spirit of the present invention.

EXPLANATION OF REFERENCES

10: ultrasound diagnostic apparatus
12: ultrasound probe
14: apparatus main body
16: monitor
18: cable
20: connector box
30: transmission and reception circuit
31: switch control circuit
32: shift register
40: transducer array
42: connection circuit
50: main transducer
50a: center-side main transducer
50b: first end-side main transducer
50c: second end-side main transducer
52: sub-transducer
52a: center-side sub-transducer
52b: first end-side sub-transducer
52c: second end-side sub-transducer
60: first circuit
60a: main pattern
60b: sub-pattern
62: second circuit
62a: cable-side pattern
62b: first end-side main pattern
62c: second end-side main pattern
62d: first end-side sub-pattern
62e: second end-side sub-pattern
70: first substrate
72: second substrate
80: first end-side block
80a: first end-side first block
80b: first end-side second block
82: second end-side block
82a: second end-side first block
82b: second end-side second block
S: signal line
SW**1**, SW**2**, SW**2-1**, SW**2-2**, SW**3**, SW**3-1**, SW**3-2**: switch
CS, CS**1**, CS**2**, CS**2-1**, CS**2-2**: control signal line
A: opening

What is claimed is:

1. An ultrasound probe comprising:

a transducer array consisting of a plurality of transducers that irradiate a subject with an ultrasonic wave and receive a reflected wave from the subject; and

a connection circuit for electrically connecting an apparatus main body of an ultrasound diagnostic apparatus and the transducer array,

wherein the transducer array includes a plurality of main transducers arranged in a major axis direction and consisting of

a plurality of center-side main transducers located on a center side of the arrangement,

a plurality of first end-side main transducers located on a first end side of the arrangement with respect to the plurality of center-side main transducers, and

a plurality of second end-side main transducers located on a second end side opposite to the first end side of the arrangement with respect to the plurality of center-side main transducers, and

the connection circuit includes

a first circuit connecting the apparatus main body and a center-side main transducer of the plurality of center-side main transducers, and

a second circuit having a plurality of main transducer switching switches for switching a connection destination of the apparatus main body between (i) a first end-side main transducer of the plurality of first end-side main transducers and (ii) a second end-side main transducer of the plurality of second end-side main transducers located on the second end side opposite to the first end side of the arrangement.

2. The ultrasound probe according to claim **1**,

wherein each of the main transducer switching switches switches the connection destination of the apparatus main body between a k-th first end-side main transducer from the first end side among the plurality of first end-side main transducers in the arrangement and a k-th second end-side main transducer from a side of the plurality of center-side main transducers among the plurality of second end-side main transducers.

3. The ultrasound probe according to claim **1**,

wherein the first circuit is provided on a first substrate, and

the second circuit is provided on a second substrate different from the first substrate.

4. The ultrasound probe according to claim **1**,

wherein the transducer array includes a plurality of sub-transducers disposed on both sides of the plurality of respective main transducers in a minor axis direction orthogonal to the major axis direction, and

the first circuit and the second circuit each include a sub-transducer switching switch for switching between connection and non-connection between a signal line connected to (a) a main transducer of the plurality of main transducers and (b) the plurality of sub-transducers disposed on both sides of the main transducer in the minor axis direction.

5. An ultrasound diagnostic apparatus comprising:

the ultrasound probe according to claim **1**;

a transmission and reception circuit that transmits a transmission signal to the ultrasound probe, and that performs signal processing on a reception signal from the ultrasound probe; and

a switch control circuit that controls switching of the main transducer switching switch, wherein in a state in which the apparatus main body and the first end-side main transducer are connected to each other, the transmission and reception circuit transmits the transmission signal to the first end-side main transducer and the plurality of center-side main transducers, and then moves an opening formed from plural main transducers of the plurality of main transducers which are targets of transmission of the transmission signal toward the second end side,

the switch control circuit switches the main transducer switching switch so that the apparatus main body and the second end-side main transducer are connected to each other, when an end of the opening on the second end side has reached an end of the center-side main transducer on the second end side, and

in a state in which the apparatus main body and the second end-side main transducer are connected to each other, the transmission and reception circuit transmits the transmission signal to the plurality of center-side main transducers and the second end-side main transducer, and then moves the opening toward the second end side.

6. The ultrasound diagnostic apparatus according to claim **5**, wherein

the plurality of first end-side main transducers are divided into a plurality of first end-side blocks, each of which is a unit for performing switching control of the main transducer switching switch, and each including plural first end-side main transducers which are successive in the arrangement,

the plurality of second end-side main transducers are divided into a plurality of second end-side blocks, each of which is a unit of switching control of the main transducer switching switch, each corresponding respectively to each of the plurality of first end-side blocks according to the arrangement, and each including plural second end-side main transducers which are successive in the arrangement,

in a state in which the apparatus main body and each first end-side block of the plurality of first end-side blocks are connected to each other, the transmission and reception circuit transmits the transmission signal to the first end-side block and a part of the plurality of center-side main transducers, and then moves an opening formed from plural main transducers of the plurality of main transducers which are targets of transmission of the trans mission signal toward the second end side,

the switch control circuit switches the main transducer switching switch so that the apparatus main body is disconnected from a first end-side first block which is the first end-side block positioned at a position closest to the first end, and the apparatus main body and a second end-side first block which is the second end-side block adjacent to the center-side main transducer are connected to each other, when an end of the opening on the second end side has reached an end of the center-side main transducer on the second end side, and

in a state in which the apparatus main body is connected to one or more of the plurality of first end-side blocks other than the first end-side first block and the second end-side first block, the transmission and reception circuit transmits the transmission signal to the one or more of the plurality of first end-side blocks other than the first end-side first block, the plurality of center-side main transducers, and the plural second end-side main transducers included in the second end-side first block, and then moves the opening toward the second end side.

7. The ultrasound diagnostic apparatus according to claim **6**, wherein

a maximum width APmax of the opening is represented by:

$$AP\mathrm{max}=USch-N$$

wherein USch=T−(B×N), wherein T represents the number of the plurality of transducers, B represents the number of the first end-side blocks or the second end-side blocks, and N represents the number of transducers included in one first end-side block or one second end-side block.

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